Finite-cell thermodynamics, path stability, and harmonic estimator diagnostics

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This supplement separates three questions from the class generator comparison: what a small finite static model predicts exactly, whether a saved migration path is numerically stable, and how a thermal estimator behaves on a known harmonic signal. Completed negative results are retained. No new DFT, MD event search, MLIP fine-tuning or physical activation free-energy calculation occurs.

1 Exact enumeration on the literal Hu 12-site parent

The stored $Li_{12}Co_{12}O_{24}$ parent is an optimized Hu frame, not an ideal symmetrized crystal. Its 144 retained Li-site permutations were accepted at 0.001 Å coordinate tolerance; the maximum species-matching residual is 0.000263895 Å. The complete combinatorial grouping has 87 orbits over all counts. At 3/6/9 present Li it has 6/17/6 orbits containing 220/924/220 raw masks. Those counts are exact for the retained permutation action, but the literal coordinate Hamiltonian need not be exactly invariant under it.

Accordingly, every one of the 1,364 raw geometries was physically evaluated with both UMA-S-1.2.1/`omat` and MACE-MH-1/`omat_pbe`, FP32/TF32 off, without relaxation. Two fixed masks per composition each received two additional evaluations: 1,376 calls per teacher, 2,752 total, 0 failures. An evaluation retains literal parent coordinates/species/cell and deletes the selected Li. A raw ID and geometry hash identify that actual geometry; an orbit key is separate. The stored historical representative mask, rather than a geometry inferred from its canonical-code ID, selects the representative approximation.

At a fixed teacher and count, let E_i be total-cell potential energy in eV, M the number of raw states, T temperature in K and $k_B = 8.617333262145 \times 10^{-5}$ eV/K. With $\beta = 1/(k_B T)$ and $E_0 = \min_i E_i$,

$$Z_{\mathrm{rel}} = \sum_{i=1}^{M} e^{-\beta(E_i - E_0)}, \qquad p_i = e^{-\beta(E_i - E_0)}/Z_{\mathrm{rel}}, \tag{1}$$

$$\langle E \rangle = \sum_i p_i E_i, \qquad S = -k_B \sum_i p_i \log p_i, \tag{2}$$

$$F = E_0 - k_B T \log Z_{\mathrm{rel}} = \langle E \rangle - TS. \tag{3}$$

p_i is dimensionless; S has eV/K per simulated cell. The normalization is separate for each composition: no lithium chemical-potential ensemble is implied. The exact-reference effective state count $1/\sum_i p_i^2$ measures concentration of this finite distribution, not reliability of an MD sample.

For group a, g_a counts its raw members and $P_a = \sum_{i \in a} p_i$ is its probability. The exact entropy identity is

$$S = -k_B \sum_a P_a \log P_a + \sum_a P_a S(i \mid a), \quad S(i \mid a) = -k_B \sum_{i \in a} \frac{p_i}{P_a} \log \frac{p_i}{P_a}. \tag{4}$$

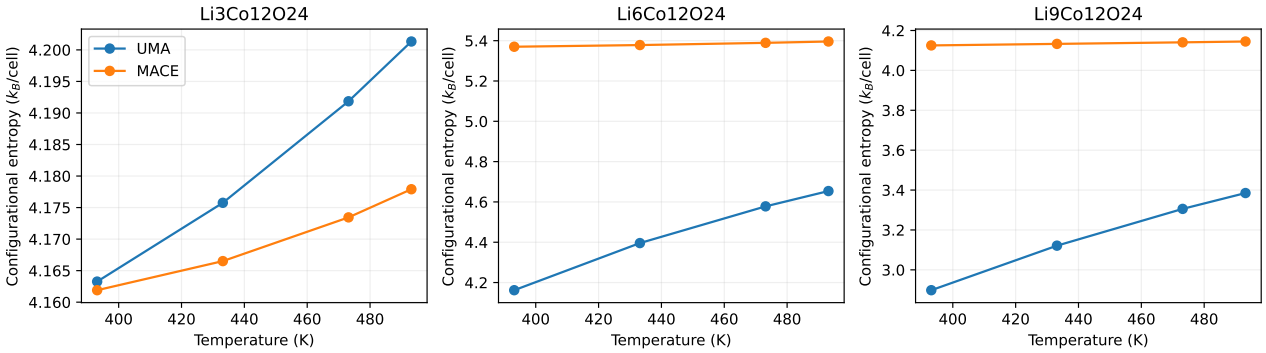
Only equal-energy members give $S(i \mid a) = k_B \log g_a$ and therefore $S = -k_B \sum_a P_a \log(P_a/g_a)$. Keeping only $-\sum P_a \log P_a$ drops within-orbit entropy. This identity was checked under an exactly symmetric null and independently on the raw data; no CVAE latent entropy is involved.

1.1 Measured approximate symmetry and teacher disagreement

The largest within-orbit energy ranges are 2.730 meV/cell for UMA and 2.777 for MACE. Maximum repeated-identical-geometry ranges are 0.00650 and 0.01526 meV/cell, respectively. Thus approximate symmetry is detectably inexact beyond repeat noise. Its effect on representative-weight thermodynamics is nevertheless small here: at 433.15 K the largest orbit-probability total variation is 0.00176. This empirical smallness is not permission to assume equality in another parent.

Li count	Raw/orbits	S_{U}/k_B	S_{M}/k_B	$N_{\mathrm{eff,U}}$	$N_{\mathrm{eff,M}}$
3	220/6	4.1758	4.1665	63.38	63.99
6	924/17	4.3958	5.3783	33.99	212.59
9	220/6	3.1213	4.1328	11.09	60.67

Here U/M denote UMA/MACE, and all values are at 433.15 K. The entropy difference at half filling is therefore mainly teacher dependence, rather than the small representative-weight approximation error. The teachers select different lowest *mean-energy orbits* at every count. Their group-mean energy rank Spearman coefficients are 0.714/0.814/0.714 at 3/6/9 Li, respectively. Raw rank coefficients are lower(approximately 0.54/0.51/0.54) and also include the ordering of nearly degenerate members inside an approximate orbit. Neither analysis identifies which teacher is closer to DFT.



Finite-reference pool errors. A uniform-raw IID pool with replacement contains L draws $I_1, \ldots, I_L$ from the known universe of M states. Its partition estimator is

$$\widehat{Z}_{\mathrm{rel}} = \frac{M}{L} \sum_{\ell=1}^{L} e^{-\beta(E_{I_\ell} - E_0)}, \quad \widehat{S} = k_B[\log \widehat{Z}_{\mathrm{rel}} + \beta(\widehat{\langle E \rangle} - E_0)]. \tag{5}$$

The normalized sample weights form $\widehat{\langle E \rangle}$. The partition estimate is unbiased under this sampling law; its logarithm and resulting energy/entropy estimates need not be. Repeated raw draws retain their particle multiplicity; adding symmetry multiplicities again would double count. Five seeds and nested 32/64/128/256/512/1,024 draw budgets quantify error, probability mass covered, bin-mass error and particle ESS. A proposal-histogram entropy would answer another question. Nested budgets share random draws and are not independent replicates. This is finite static-model validation only; it neither validates a 48-site model on 12 sites nor estimates real-LCO entropy.

2 NEB: fixed endpoints, route initialization and numerical stability

The nudged elastic band(NEB) method connects given endpoints using a chain of interior atomistic images. Physical forces perpendicular to the chain relax the route, while artificial springs distribute images. A climbing image reverses the force component along the local path tangent to seek a maximum along that direction [1]. Its projected-force stopping criterion differs from the physical force at ordinary nonclimbing images; those images are not expected to be individual local minima.

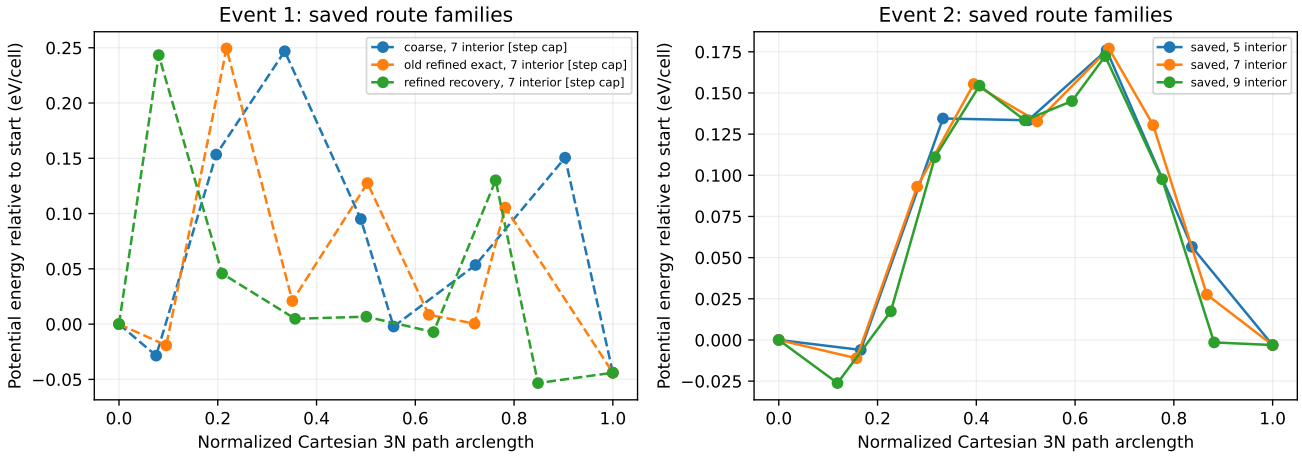
Two saved MD events retain continuous atom identities and periodic branches. The marked event 1 Li is local atom 65(parent 66), moving between parent Li sites 18 and 31; event 2 is local/parent atom 59, between sites 11 and 23 with periodic image $(1, 0, 0)$. The ideal-site displacement in each is 2.8113Å. Independent canonicalization of endpoints would destroy this event information; an event can be real even if its endpoint occupation orbits are symmetry-equivalent.

The first tranche held the exact old endpoint coordinates and cell fixed, resampled the saved route families to seven interior images, and used the common 0.02 eV/Å projected-force target. ASE 3.26.0, improved tangent, climbing image, $k = 0.1$ eV/Å^2, FIRE $dt = 0.1$, $dt_{\max} = 1$, maximum step 0.05Å, serial images sharing one GPU backend through independent caches, and no new IDPP/event search were declared. Each band is capped at 500 optimizer steps, 45 minutes including imports/loading, and 5,000 actual E/F/stress calls.

Case	Steps	Projected max force eV/Å	Terminal forward span eV/hop
Event 1 coarse route, 7 interior	500	0.2253	0.2468(not converged)
Event 1 refined route, 7 interior	430+interruption	0.8877	not a barrier estimate
Event 1 bounded recovery	69	0.4929	0.2433(not converged)
Event 1 old 7 exact coordinates	500	0.3471	0.2493(not converged)
Event 2,5 interior	11	0.01338	0.17598
Event 2,7 interior	21	0.01997	0.17702
Event 2,9 interior	33	0.01999	0.17237

The original event 1 refined worker was externally killed by Slurm fail-fast when its sibling reached the scientific step cap. Its 430-step snapshot and 3,020 successful calls are preserved. One additional in-flight position update is conservatively charged: only 69 of the original 500 steps remain. Recovery restores that exact snapshot, resets FIRE memory explicitly, and preserves the original time/call reservation; it is not a fresh 500-step rerun. Equal-arclength resampling redistributes even an already-seven-image band, so an additional declared direct-coordinate continuation tested that initialization difference. It too hit the 500-step cap. Resampling alone therefore does not explain the failure.

The event 2 forward-span spread across 5/7/9 interior images is 0.00465 eV/hop, below the predeclared 0.02 eV/hop practical numerical target. This is a chosen numerical criterion, not model/DFT accuracy. Event 1 remains unresolved; convenient terminal spans must not be selected as validated barriers. Event 2 endpoint physical forces remain 0.02954/0.01264 eV/Å because the endpoints were held fixed. Numerical image stability is conditional on those endpoints and that potential.



Dashed profiles above are preserved terminal states of nonconverged event 1 calculations. The horizontal coordinate is normalized full-3N Cartesian arclength on the saved unwrapped branch, not a measured reaction time.

2.1 A stable local unstable mode does not verify the complete hop

The fixed-seven-image continuation reached the 0.01 eV/Å projected-force target in 82 steps, with forward/reverse spans 0.16918/0.17231 eV. The forward change from the 0.02 target is −0.00784 eV, within the declared numerical criterion. The candidate climbing image 5 has physical maximum force 0.00888 eV/Å.

Two central finite-difference Hessians use the *same exact candidate* at $\delta = 0.01$ and 0.005Å, with three fresh identical-coordinate force calls before each. For 180 atoms, each full Cartesian Hessian costs 1,080 displaced force calls. With $F_a = -\partial E/\partial R_a$,

$$H_{ab} \simeq -\frac{F_a(R + \delta e_b) - F_a(R - \delta e_b)}{2\delta}. \tag{6}$$

H is in eV/Å^2. The mass-weighted matrix is $D = M^{-1/2} H M^{-1/2}$, with atomic masses in amu. After symmetrization and removal of exactly three translations, both 537-dimensional internal spectra have exactly one negative eigenvalue: $-0.111003/-0.111010$ eV Å^{-2} amu^{-1}. The next is +0.003343; the two negative-mode vectors have absolute overlap 0.999999994. Observed force-component repeat ranges are about 2.2×10^{-6} eV/Å, but these three-call ranges are not calibrated Hessian error bars. Independently reconstructing both Hessians from the raw force pairs reproduces all eigenvalues; all 2,160 displacement hashes and logged force maxima were checked. Unweighted force-derivative noise and mass-weighted eigenvalues have different units; their magnitudes are not directly compared. Displacement-size agreement and mode overlap are the actual numerical evidence.

A negative curvature means displacement along that local mode lowers energy to second order when the gradient is small. It does not identify the basins reached farther away. The negative mode's absolute overlap with the actual ASE improved tangent is 0.779 in the mass metric, or 0.920 when both are represented as normalized Cartesian displacements. Those are different coordinate metrics, not probabilities or independent physical replications.

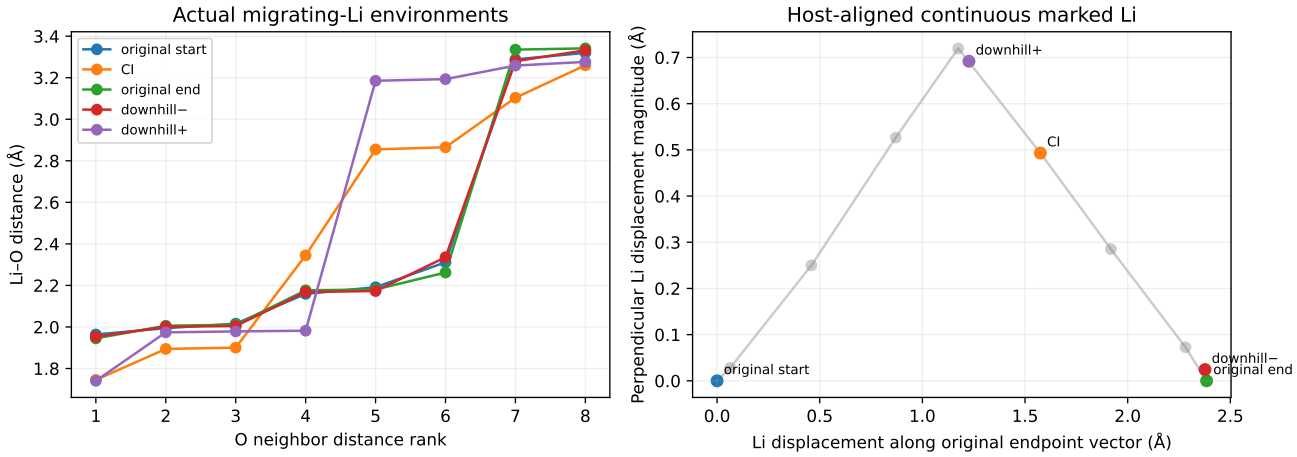
These calculations produce MLIP potential-energy path spans in eV per marked hop. They establish neither an activation free energy nor diffusivity. Later physical validation needs endpoint/saddle/path DFT under compatible electronic and spin settings. No VASP job or input deck is submitted during this goal.

2.2 Same raw vacancy mask, different continuous configurations

The plus/minus negative-mode displacements were independently relaxed at a 0.02 eV/Å maximum-force target. The minus branch is end-like: its identity-preserving, host-aligned RMS difference from the original end is 0.04686Å, with maximum atomic difference 0.06628Å. The plus branch instead reaches an off-site, force-small state: RMS 0.11671Å and maximum difference 1.34843Å from that same end. Its migrating Li is 1.5992Å from its nearest parent site, with only 0.05791Å separating the first/second site distances.

State	E (eV/cell)	Li/site distance Å	4th O distance Å	5th O distance Å
Original end	−1032.516551	0.2562	2.1763	2.1806
Climbing image	−1032.344237	1.1984	2.3447	2.8548
Downhill minus	−1032.521675	0.2735	2.1684	2.1729
Downhill plus	−1032.378428	1.5992	1.9820	3.1853

The original end, climbing image, minus state and plus state all receive the *same raw nearest/Hungarian vacancy assignment*, hence the same orbit key, despite these geometric and energy differences. The plus state has four nearer O atoms, all within 1.982Å, followed by a 1.203Å gap to its fifth O; the minus state has six O within 2.336Å. These ranked distances require no unvalidated bonding cutoff. Full periodic neighbors were searched within 5Å only to obtain the distance lists.



The right panel shows the actual continuous marked-Li geometry after removing common host translation. The grey chain is the saved NEB route; perpendicular displacement is a magnitude, not a signed plane coordinate. The off-site endpoint is near an intermediate part of that route. This supports a possible compound path with an intermediate state; neither its minimum character nor the complete sequence of elementary connections has been certified. It does not invalidate a protocol-defined pristine-mask-to-endpoint dataset, but it prevents using a vacancy orbit as a unique continuous PES-basin identity.

On the unchanged original seven-image UMA path, MACE gives forward/reverse spans 0.16516/0.25159 eV and selects peak image 3 rather than UMA's 5. MACE endpoint physical forces are 0.2567/0.2547 eV/Å: these are not stationary MACE endpoints, so the profile is not a MACE-optimized barrier. Independently tightening the same original pair under each teacher defines a separate new-endpoint experiment.

2.3 Endpoint sensitivity prevents an overall stable-barrier claim

Both original endpoints were independently tightened to 0.01 eV/Å under each teacher, retaining the marked Li assignments. UMA lowered their energies by 0.02151/0.000034 eV; MACE lowered them by 0.13892/0.10486 eV. Each new band kept seven interior images and used the 0.02 eV/Å target. Initial interiors were the original path plus a linear endpoint-displacement taper in its original Cartesian arclength, with no quiet image resampling or new IDPP.

New endpoints	Forward span eV	Reverse span eV	End minus start eV	Peak image
UMA	0.195626	0.177294	+0.01833	5
MACE	0.199585	0.251953	−0.05237	3

The original fixed-seven-image 0.01 case gave 0.169182 eV forward, while the new UMA-endpoint band gives 0.195626 eV: a +0.026444 eV change, exceeding the chosen 0.02 eV practical numerical target. Both endpoint coordinates and the band force target differ in this comparison; it measures combined protocol sensitivity, not an isolated endpoint-error contribution. The restricted fixed-original image/tolerance comparison passed, but *overall barrier stability is not established*. A small force residual alone supplies no energy error bound without information about local curvature.

The similar UMA/MACE forward heights are also insufficient: the reaction-energy sign and highest-image identity differ, and the original marked path has not been verified as one elementary hop. These are MLIP potential-energy path spans, not validated activation free energies. No additional physics is run after this bounded study; later DFT and fuller basin/connection verification remain necessary.

3 Actual 537-mode harmonic null for the thermal estimator

The earlier thermal work had a 21-mode synthetic check. The new check uses the actual saved positive 537-mode spectrum and mass-weighted eigenvectors from one 180-atom basin. Three independent phase seeds generate nested 10/20/50 ps synthetic velocity observations at 0.5 fs; each is analyzed by pinned ATLAS py-xPT 2.1.0, commit`197bdcf5f57c9d04b5527f9e7d6f4fdb8bf3b82d`, under 1PT/2PT/3PT. All 27 analyses completed without failures. An allocation auto-logout interrupted the first attempt; its four complete results were reused immutably and its incomplete raw output was preserved. No MLIP or MD calls were required.

For mode j, frequency ν_j in THz(cycles/ps), normalized mass-weighted eigenvector Q_j and uniform phase ϕ_j,

$$\dot{q}_j(t) = \sqrt{2k_BT/C}\cos(2\pi\nu_j t + \phi_j), \qquad v(t) = M^{-1/2}Q\dot{q}(t). \tag{7}$$

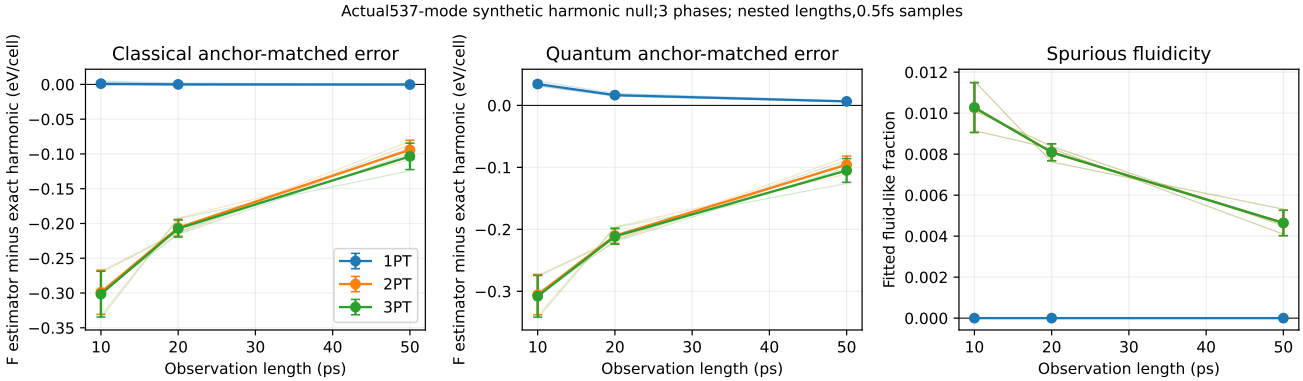
M is the diagonal mass matrix in amu; $C = 1.0364269656262175 \times 10^{-4}$ converts amu $\text{Å}^2/\text{ps}^2$ into eV. Every mode has fixed total energy k_BT. Random phases match canonical harmonic velocity second moments, but fixed modal energies are not a canonical MD trajectory. The known harmonic Hamiltonian nevertheless supplies an exact partition-function reference. The energy anchor is $E_{\min} + 537k_BT$ and is included once. Only 3 COM modes are excluded; no separate molecular rotations/translations are subtracted from the crystal.

Finite windows have a slightly shifted observed kinetic temperature. We compare entropy at that native temperature and explicitly match the supplied total-energy anchor rather than interpreting a first-order temperature transport as exact. If $\mathcal{E}$ is that anchor and S_h, U_h are exact harmonic quantities,

$$F_{\text{c,ref}} = \mathcal{E} - TS_{\text{c,h}}, \tag{8}$$

$$F_{\text{q,ref}} = \mathcal{E} + [U_{\text{q,h}} - (E_{\min} + 537k_BT)] - TS_{\text{q,h}}. \tag{9}$$

At the target temperature these reduce to the usual classical/quantum harmonic free energies. At the observed temperature they isolate estimator error from the known finite-window anchor mismatch.



The mean 1PT quantum anchor-matched error decreases from about+0.0344 eV/cell at 10 ps to+0.00645 at 50 ps. In contrast, 2PT/3PT infer nonzero fluidicity 0.0041–0.0116 from an analytically bounded harmonic signal with no diffusion. At 50 ps, 3PTquantum errors range from−0.0884 to−0.1260eV/cell across phases. Longer windows help, but do not establish convergence at these lengths.

COM speeds are numerically negligible; energy-anchor round trips are exact, $F = U - TS$ residuals are at floating-point roundoff, and the integrated spectrum contains 537 modes to within about1.2×10^{-6}modes. Correct mode normalization alone therefore does not imply a correct free-energy estimate. Phase spread is descriptive, and nested lengths are correlated. This is a harmonic estimator null, not an anharmonic correction, an unrestricted equilibrium MD result or validation of real-LCO thermodynamics. Discrete vacancy entropy and within-basin vibrational entropy cannot simply be added without a consistent basin partition, energy convention and statistical measure.

Evidence, execution and later physical validation

Raw small-cell labels, per-geometry hashes, stored representative masks, teacher/precision receipts and finite-model tables are in `reports/overnight_20261108/smallcell`. Independent arithmetic and literal geometry checks are in`smallcell_independent_v2`. A first checker failure was retained: its roundoff scale incorrectly used a small energy difference rather than the magnitudes of the subtracted energies. The repaired arithmetic bound uses the same 128 machine-epsilon factor with the operand scale; no scientific label or result changed. The independent check verifies 1,364 geometries and 11,104 numeric fields, with maximum difference2.84×10^{-13}.

NEB inputs, full physical/projected forces, continuous parent atom IDs, unconverged routes, actual call logs and scheduler interruption evidence are retained. Harmonic raw spectra/configuration/native results, deterministic phases, input Hessian/minimum hashes and the reused-result lineage remain available. The new condition schemas explicitly state or link authoritative FP32/TF32-off inference receipts; old-schema omissions are not treated as evidence of wrong inference.

The later-DFT geometry handoff contains the candidate, both original endpoints, both downhill states and every intermediate band image as exact-coordinate NPZ with original same-geometry UMA energy/force/stress labels, plus POSCAR/CIF viewing exports. Local-to-parent atom IDs, source/checkpoint hashes and coordinate hashes are preserved. The off-site plus state is explicitly retained as an example of the limitations of a vacancy-only representation; it is not silently merged with the end-like state sharing its assigned mask.

Exact commands and final allocation/cost/publication receipts belong to the integrated goal packet. New code is under `scripts/overnight_20261108`; no new allocation is requested by these analysis scripts. Model weights are not redistributed. Resource accounting distinguishes allocated node time, GPU-device time, actual static E/F calls, relaxations and analysis work; final totals are attached after job release.

Ricardo should inspect physical paths, endpoint residuals and teacher disagreement before selecting later DFT candidates. Beihao's class task remains independent model-mechanism verification, not demonstrating a physical entropy/barrier claim. Varad should independently replay raw probabilities, multiplicities, repeated queries, failed attempts and costs. A stable same-MLIP result still needs later DFT and experimental comparison before a physical recycling conclusion.

References

[1] Atomic Simulation Environment developers. Nudged elastic band: ASE documentation, 2026. Installed implementation3.26.0; method and optimizer source hashes retained.